

Substrate-Symplectic Content Across Integer-Weight K-Types V_(1,0), V_(1,1), V_(2,0) v0.1

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Substrate-Symplectic Integer-Weight K-Types v0.1

0. Goal

Per Casey “queue never empties” + Cal #36 STANDING positive-search + Substrate-Symplectic v0.2 Sp(2) Hamiltonian flow + Thursday Afternoon Synthesis v0.1 § “exploration of substrate-symplectic substrate-mechanism across other K-types”: substrate-symplectic substrate-mechanism reach across integer-weight K-types V_(1, 0), V_(1, 1), V_(2, 0) via Sp(2) substrate-symplectic Lie group action.

1. Sp(2) Substrate-Symplectic Representations

Per Sp(2) = Sp(2, $\mathbb{C}$) rank-2 symplectic Lie group representation theory:

Sp(2) representation	dim	Substrate K-type cross-link via Spin(5) $\cong$ Sp(2)
Trivial	1	V_(0, 0) (substrate Higgs vacuum)
Fundamental	4	V_(1/2, 1/2) (substrate-spinor; Spin(5) double-cover content)
Adjoint	10	V_(1, 1) (substrate-spectral so(5) adjoint = substrate-symplectic sp(2) adjoint)
Symmetric Sym ² of fund	10	V_(1, 1) or V_(2, 0) decomp
Λ^2 of fund	5	V_(1, 0) (substrate-vector; substrate-symplectic Λ^2 antisymmetric)
Sym ² traceless	14	V_(2, 0) (substrate-spin-2; substrate-symplectic Sym ² traceless)

Substrate-mechanism observation: Sp(2) substrate-symplectic representations EQUIVALENT to substrate K-types on D_IV⁵ via Spin(5) $\cong$ Sp(2) accidental isomorphism + standard Lie-group representation theory cross-link.

Substantive cross-link to SSG-1, 2, 3, 11, 13: - SSG-1 V_(1/2, 1/2) $\leftrightarrow$ Sp(2) fundamental (substrate-spinor) - SSG-2 V_(1, 1) $\leftrightarrow$ Sp(2) adjoint (substrate-Lie-algebra dim 10) - SSG-3 V_(1, 0) $\otimes$ V_(1, 0) decomp $\leftrightarrow$ Sp(2) tensor product Fund $\otimes$ Fund - SSG-11 V_(2, 0) $\leftrightarrow$ Sp(2) Sym² traceless - SSG-13 V_(0, 0) $\leftrightarrow$ Sp(2) trivial

2. Substrate-Symplectic Content at Integer K-Types

$V_{-}(1, 0)$ Substrate-Vector $\leftrightarrow \text{Sp}(2) \Lambda^2$ Fundamental

Per accidental isomorphism + standard Lie-group rep theory: $V_{-}(1, 0)$ substrate-vector dim 5 corresponds to Λ^2 (antisymmetric 2-product) of $\text{Sp}(2)$ fundamental.

Substrate-symplectic content: $V_{-}(1, 0)$ carries substrate-symplectic 2-form structure as antisymmetric pair of substrate-fundamental $V_{-}(1/2, 1/2)$ elements.

Substantive substrate-mechanism reading: substrate-photon K-type $V_{-}(1, 0)$ (gauge potential A_{μ}) has substrate-symplectic substantive content via substrate-spinor pair antisymmetric composition — substrate-gauge potential as substrate-symplectic 2-form derivative.

$V_{-}(1, 1)$ Substrate-Adjoint $\leftrightarrow \text{Sp}(2)$ Adjoint

Per accidental isomorphism + standard Lie-group rep theory: $V_{-}(1, 1)$ substrate- $\text{so}(5)$ adjoint dim 10 corresponds to $\text{Sp}(2)$ adjoint representation.

Substrate-symplectic content: $V_{-}(1, 1)$ carries substrate-Hamiltonian flow generators (substrate-symplectic adjoint $\text{sp}(2)$ algebra).

Substantive substrate-mechanism reading: substrate- $F^{\mu\nu}$ gauge field strength (SSG-2) $\leftrightarrow$ substrate-symplectic Hamiltonian generator. Substrate-EM field substrate-Hamiltonian flow on substrate-photon K-type $V_{-}(1, 0)$.

$V_{-}(2, 0)$ Substrate- Sym^2 -Traceless $\leftrightarrow \text{Sp}(2) \text{Sym}^2$ Traceless

Per accidental isomorphism + standard Lie-group rep theory: $V_{-}(2, 0)$ substrate- Sym^2 -traceless dim 14 corresponds to $\text{Sp}(2) \text{Sym}^2$ traceless representation.

Substrate-symplectic content: $V_{-}(2, 0)$ carries substrate-stress-energy substrate-symplectic content + substrate-graviton-sector substrate-symplectic structure (SSG-11 v0.9 Wednesday).

Substantive substrate-mechanism reading: substrate-stress-energy $T_{\mu\nu}$ (Einstein equation framework v0.1) + substrate-graviton $h_{\mu\nu}$ substrate-symplectic content via $\text{Sp}(2)$ substrate-symplectic Sym^2 substrate-mechanism.

3. Substrate-Symplectic Cat 6 UNIVERSAL Across Substrate K-Type Spectrum

Per $\text{Sp}(2)$ substrate-symplectic Lie group action: substrate-symplectic substrate-mechanism is UNIVERSAL across substrate K-type spectrum ($V_{-}(0, 0)$, $V_{-}(1/2, 1/2)$, $V_{-}(1, 0)$, $V_{-}(1, 1)$, $V_{-}(2, 0)$, $V_{-}(3/2, 1/2)$, $V_{-}(5/2, 1/2)$, ...).

Substantive substrate-mechanism observation: Cat 6 substrate-symplectic substrate-mechanism is NOT confined to substrate-spinor (half-integer $V_{-}(\lambda_{-}1, 1/2)$) K-types; it extends across ALL substrate K-types via $\text{Sp}(2)$ substrate-symplectic representations.

Substrate-Pfaffian $\text{Pf}(\omega)$ content per K-type: - $V_{-}(0, 0)$ trivial: $\text{Pf}(\omega) = 1$ (substrate-vacuum) - $V_{-}(1/2, 1/2)$ fundamental 4-dim: $\text{Pf}(\omega) = 1$ substrate-natural ($\text{Sp}(2)$ substrate-symplectic standard normalization) - $V_{-}(1, 0) \Lambda^2$ 5-dim: $\text{Pf}(\omega)$ substrate-natural — substrate-vector substrate-symplectic content - $V_{-}(1, 1)$ adjoint 10-dim: $\text{Pf}(\omega)$ substrate-natural — substrate-adjoint substrate-symplectic Hamiltonian generators - $V_{-}(2, 0) \text{Sym}^2$ traceless 14-dim: $\text{Pf}(\omega)$ substrate-natural — substrate- Sym^2 substrate-symplectic content

4. Cross-Link to Gate 4 v0.2 Substrate-Pfaffian Resolution Candidate

Per Gate 4 v0.2 (Thursday ~12:02 EDT): substrate-symplectic substrate-Pfaffian $1/g$ substrate-natural factor resolves naive composition factor $\sim 7 \rightarrow 1432/g = 204.6 \approx T190$ at Tier 2 STRUCTURAL.

Substantive substrate-mechanism extension: substrate-Pfaffian factor $1/g$ substrate-natural may be UNIVERSAL across substrate K-types via $\text{Sp}(2)$ substrate-symplectic structure — not specific to gen-2 $V_{-}(3/2, 1/2)$ muon K-type.

Cross-K-type substrate-mechanism prediction: substrate-symplectic substrate-Pfaffian $1/g$ substrate-natural factor enters substrate-mass / substrate-coupling observables across substrate K-type spectrum, contributing substrate-natural normalization to Mehler kernel composition at substrate-symplectic level.

Multi-week explicit verification: substrate-Pfaffian $\text{Pf}(\omega)$ values at substrate K-types $V_{-}(0, 0)$, $V_{-}(1/2, 1/2)$, $V_{-}(1, 0)$, $V_{-}(1, 1)$, $V_{-}(2, 0)$, $V_{-}(3/2, 1/2)$, $V_{-}(5/2, 1/2)$; substrate-symplectic substrate-natural factor per K-type per $\text{Sp}(2)$ substrate-symplectic structure.

5. Substrate-Hamiltonian Flow at Integer K-Types

Per $\text{Sp}(2)$ substrate-symplectic Hamiltonian flow: substrate-Hamiltonian $H_B = \sqrt{H_B^2}$ acts on substrate K-types via $\text{Sp}(2)$ substrate-symplectic flow.

Substrate-time evolution $\rho_{\text{commit}}(\tau)$ (Tier 0 v0.1.6) decomposes via substrate-symplectic Hamiltonian flow:
- $V_{-}(0, 0)$: substrate-vacuum substrate-time evolution (substrate-Higgs sector) - $V_{-}(1/2, 1/2)$: substrate-spinor substrate-Hamiltonian flow (Class A substrate-Casimir + chirality projection) - $V_{-}(1, 0)$: substrate-gauge potential substrate-Hamiltonian flow (substrate-Maxwell) - $V_{-}(1, 1)$: substrate-gauge field strength substrate-Hamiltonian flow (substrate-EM) - $V_{-}(2, 0)$: substrate-stress-energy substrate-Hamiltonian flow (substrate-Einstein eq)

Substrate-symplectic substrate-mechanism unifies substrate-time evolution across substrate K-type spectrum.

6. Closure v0.1

Substrate-Symplectic Integer-Weight K-Types v0.1: substrate-symplectic Cat 6 substrate-mechanism UNIVERSAL across substrate K-type spectrum ($V_{-}(0, 0)$, $V_{-}(1/2, 1/2)$, $V_{-}(1, 0)$, $V_{-}(1, 1)$, $V_{-}(2, 0)$, $V_{-}(3/2, 1/2)$, $V_{-}(5/2, 1/2)$).

Substantive cross-links: - SSG-1, 2, 3, 11, 13 $\leftrightarrow$ $\text{Sp}(2)$ substrate-symplectic representations - Substrate-photon $V_{-}(1, 0)$ carries substrate-symplectic Λ^2 content - Substrate-EM field $V_{-}(1, 1)$ carries substrate-Hamiltonian generators - Substrate-stress-energy $V_{-}(2, 0)$ carries substrate-Sym² substrate-symplectic content - Substrate-time evolution $\rho_{\text{commit}}(\tau)$ = substrate-symplectic Hamiltonian flow across all K-types

Multi-week explicit substrate-Pfaffian per K-type derivation gates substrate-symplectic substrate-mechanism FORCING form per Cal #189.

Substantive substrate-mechanism cascade implication: Cat 6 substrate-symplectic may be UNIVERSAL substrate-mechanism layer across substrate observables — contributing substrate-natural normalization to Mehler kernel composition at substrate K-type spectrum level.

— Lyra, Thu 2026-06-04 ~12:15 EDT. Substrate-Symplectic Integer K-Types v0.1: Cat 6 substrate-symplectic substrate-mechanism UNIVERSAL across substrate K-type spectrum via $\text{Sp}(2) \cong \text{Spin}(5)$ accidental isomorphism; substrate-Pfaffian per K-type substrate-natural; substrate-Hamiltonian flow unifies substrate-time evolution across substrate K-types; multi-week explicit per-K-type substrate-Pfaffian derivation gates substrate-mechanism FORCING form.